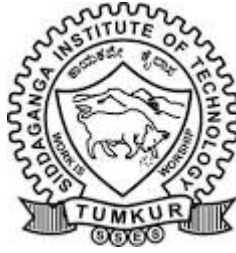


SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103
(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)



Internship Report on

“Compl-AI - an AI-powered security compliance platform”

submitted in partial fulfillment of the requirement for the award of the
degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

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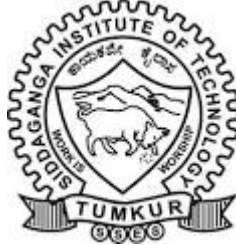
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

2025-26

SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

Electronics and Communication Engineering



CERTIFICATE

Certified that the internship work entitled “**COMPL-AI - AN AI POWERED SECURITY COMPLIANCE PLATFORM**” is a bonafide work carried out by Dharshan A (1SI22EC025) in partial fulfillment for the award of degree of Bachelor of Engineering in **Electronics and Communication Engineering** from Siddaganga Institute of Technology, an autonomous institute under Visvesvaraya Technological University, Belagavi during the academic year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Internship report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the Bachelor of Engineering degree.

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I offer pranams at the lotus feet of **His Holiness, Dr. Sree Sree Sivakumara Swamigalu**, Founder President and **His Holiness, Sree Sree Siddalinga Swamigalu**, President, Sree Siddaganga Education Society, Sree Siddaganga Math for bestowing upon their blessings.

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Dharshan A (1SI22EC025)

Course Outcomes

After successful completion of internship, graduates will be able to:

CO1: Formulate the given engineering problem and review literature.

CO2: Design a methodology using engineering knowledge and analytical abilities .

CO3: Implement the methodology and propose a meaningful solution by using problem solving skills considering health, safety issues.

CO4: Gain practical/research knowledge to existing classroom knowledge, such as new tools or skills .

CO5: Effectively communicate the solution through report and oral presentation.

CO6: Demonstrate time management, punctuality and life-long learning skills.

CO7: Expand the professional network and explore career options.

CO8: Demonstrate individual and team work, ethics required for success in a professional work environment.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO-1	3	3										3	
CO-2	3	2	2									3	
CO-3	3	3		2		3					3	3	3
CO-4					3						3		3
CO-5									3				3
CO-6										3	3		3
CO-7										3	3		3
CO-8							3	3					3
Average	3	3	2	2	3	3	3	3	3	3	3	3	3

Attainment level:- 1: Slight(low) 2: Moderate(medium) 3: Substantial(high)

POs: PO1: Engineering knowledge, PO2: Problem analysis, PO3: Design of solutions, PO4: Conduct investigations of complex problems, PO5: Engineering tool usage, PO6: Engineer and the world, PO7: Ethics, PO8: Individual and collaborative work, PO9: Communication, PO10: Project management and finance, PO11: Life-long learning.

Abstract

This internship was undertaken at Accenture (BDC14, Bengaluru) with the aim of gaining practical exposure to real-world applications of AI in cybersecurity compliance. With the increasing complexity of regulatory standards and the manual effort required for compliance verification, there is a growing need for intelligent automation. This internship focused on addressing these challenges by exploring AI-driven solutions for security control validation, evidence analysis, and policy management within an enterprise environment.

The primary objective of the work was to design and develop CompliAI, an intelligent SaaS platform capable of automating security compliance processes. The system is designed to ingest structured compliance control data through CSV templates and validate these controls using evidence documents such as PDFs, DOCX files, images, and plain text. A key goal was to leverage Large Language Models (LLMs) to analyze extracted content and determine compliance status (Compliant or Non-Compliant) along with meaningful explanations. Additionally, the platform aims to support automated policy generation and validation against predefined compliance standards.

The implementation involved building a full-stack application with a Flask-based RESTful backend and a Vue 3 frontend. User authentication features such as registration, login, and session management were implemented, along with MySQL database integration for managing user data and document metadata. The frontend enables CSV upload, parsing, and handling of multiple evidence files per control. A text extraction pipeline was developed using Azure Blob Storage for file handling, Azure Computer Vision (OCR) for image-based text extraction, PyMuPDF for processing PDFs, and standard libraries for DOCX and TXT parsing. The LLM pipeline was implemented using GPT-3.5 integrated through LangChain to perform compliance evaluation and generate explanatory summaries.

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Chapter 1

Introduction

Cybersecurity compliance has evolved from a best practice into a critical operational requirement for modern organizations, especially those handling sensitive and regulated data across industries such as healthcare, finance, and government. Regulatory frameworks like HIPAA, NIST Cybersecurity Framework, and SOC 2 define extensive sets of security controls that organizations must implement, monitor, and continuously validate. These controls span areas such as data protection, access management, incident response, and system integrity. Ensuring compliance with these frameworks is not a one-time task but an ongoing process that requires periodic audits, documentation, and evidence validation.

Traditionally, this compliance verification process is carried out manually by Governance, Risk, and Compliance (GRC) analysts. They review large volumes of evidence documents—such as logs, policy files, access records, and system configurations—and map them against predefined control requirements. This approach is inherently time-consuming, labor-intensive, and prone to human error. As organizations scale and manage multiple frameworks simultaneously, the complexity increases significantly, making manual compliance verification inefficient and difficult to sustain.

To address these challenges, CompliAI was developed as an AI-powered cybersecurity compliance platform that automates the control testing and evidence verification workflow. The platform allows organizations to upload their compliance controls in a structured CSV format and associate relevant supporting evidence for each control. Leveraging a Large Language Model (LLM)-based pipeline, the system intelligently analyzes the submitted evidence, evaluates whether specific control requirements are met, and generates a compliance verdict. In addition, it produces clear, human-readable summaries that explain the reasoning behind each decision, improving transparency and audit readiness. This internship project was undertaken at Accenture BDC14, Bengaluru, as part of the Security Managed Services practice. The work reflects real-world enterprise scenarios where security teams must manage GRC programs across multiple clients, regulatory

standards, and business environments. The project emphasizes scalability, automation, and accuracy-key requirements for modern compliance systems.

Through this internship, significant hands-on experience was gained in designing and developing production-grade, cloud-integrated web applications with AI at their core. The work involved integrating backend systems, handling structured and unstructured data, implementing LLM pipelines, and building user-friendly interfaces. Overall, the project demonstrates how artificial intelligence can transform traditional compliance processes into efficient, scalable, and intelligent systems capable of meeting the demands of modern cybersecurity governance.

1.1 Objectives of the internship

The internship at Accenture BDC14, Bengaluru, was undertaken to gain hands-on experience in building AI-integrated full-stack applications applied to the cybersecurity compliance domain. The following are the core objectives:

- To design the Compl-AI platform architecture.
- To implement user authentication.
- To build the CSV upload and parsing pipeline.
- To develop the text extraction pipeline.
- To integrate the LLM compliance verification engine.

1.2 Organisation of the Report

This internship report is organized into several chapters, each focusing on a specific aspect of the work carried out during the internship. Chapter 1 presents the introduction, which provides an overview of cybersecurity compliance, the motivation for the work, and the objectives of the internship. Chapter 2 describes the company/organization profile, highlighting the background, services, and role of Accenture BDC14 in cybersecurity and compliance. Chapter 3 covers the literature survey and tools & technologies used in the project, including frontend, backend, cloud services, and database technologies. Chapter 4 presents the internship description, detailing the project undertaken and the work responsibilities assigned during the internship. Chapter 5 discusses the reflections on the

internship experience, including skill development, challenges faced, and lessons learned. Chapter 6 presents the results, including system evaluation, performance metrics, experimental setup, analysis, and key findings. Finally, the report concludes with bibliography and appendices, which provide references and additional supporting information.

Chapter 2

Company/Organization Profile



Figure 2.1: Accenture Logo

Figure 2.1 shows the Accenture logo. Accenture is one of the world's leading professional services companies, delivering services in strategy, consulting, digital transformation, technology, and operations. Headquartered in Dublin, Ireland, and listed on the New York Stock Exchange, Accenture operates in more than 120 countries with a workforce of approximately 800,000 professionals worldwide. The company serves a diverse range of industries, including financial services, healthcare, telecommunications, retail, manufacturing, defence, and government, helping organizations improve performance, enhance security, and drive innovation through technology.

Accenture is widely recognized for its strong expertise in digital transformation and cybersecurity, enabling enterprises to adopt secure, scalable, and resilient solutions. The company provides end-to-end services, including risk assessment, security architecture design, compliance implementation, cloud security, and managed security services, ensuring alignment with global standards and regulatory requirements.

The Accenture Bengaluru Delivery Centre (BDC14), located in Koramangala Industrial Layout, Bengaluru, is one of the company's important delivery hubs in India. It hosts multiple specialized domains such as cybersecurity, cloud computing, data analytics, and application development. The cybersecurity practice at BDC14 plays a critical role in delivering comprehensive security solutions to global clients, covering areas such as governance, risk and compliance (GRC), threat detection and response, vulnerability management, and secure system design.

Through its advanced infrastructure and expert workforce, Accenture BDC14 supports large-scale enterprise engagements and provides a platform for practical exposure to real-world cybersecurity challenges, making it an ideal environment for developing industry-relevant technical and professional skills.

2.1 Services and Capabilities

Accenture's service portfolio spans five broad domains, enabling end-to-end digital transformation and technology-driven business solutions:

Strategy and Consulting: Business strategy, operating model design, digital transformation roadmaps, and industry-specific advisory services to help organizations navigate change and drive growth.

Technology: Cloud migration, application modernization, enterprise platform implementation (SAP, Oracle, Salesforce), infrastructure services, and engineering solutions across all major technology stacks.

Operations: Intelligent business process services, finance and accounting operations, supply chain management, and customer operations powered by automation and AI.

Cybersecurity: End-to-end security services including risk assessment, security architecture, compliance implementation (HIPAA, NIST CSF, SOC 2, ISO 27001), managed detection and response, cloud security, and identity and access management.

Industry X: Digital engineering and manufacturing services, product lifecycle management, and intelligent asset management for industrial and manufacturing clients.

2.2 Certifications and Quality Standards

Accenture adheres to globally recognized quality, security, and compliance standards across its service delivery operations:

- ISO 9001:2015 - Quality Management System
- ISO 27001:2022 - Information Security Management System
- ISO 22301:2019 - Business Continuity Management System
- SOC 2 Type II - Security, Availability, and Confidentiality Controls
- CMMI Level 5 - Software and Services Development Process Maturity

- PCI DSS Compliance - Payment Card Industry Data Security Standard
- GDPR Compliance - General Data Protection Regulation
- FedRAMP Authorization - Cloud services for US Federal Government

Chapter 3

Tools and Technologies

3.1 Frontend Technologies

Axios is a promise-based HTTP client for JavaScript. It was used to handle all communication between the Vue 3 frontend and the Flask-RESTX backend, including multipart form data requests for file uploads and JSON requests for authentication and result retrieval.

3.2 Backend Technologies

3.2.1 Python

Python serves as the primary backend language for CompliAI. Its rich ecosystem of libraries for file processing, HTTP handling, database interaction, and LLM integration made it the natural choice for the server-side implementation.

3.2.2 Flask-RESTX

Flask is a lightweight WSGI web application framework for Python. Flask-RESTX extends Flask with tools for building structured REST APIs, including automatic Swagger UI documentation, request parsing, and response marshalling. The backend API was organized into namespaces for authentication, file upload, text extraction, and compliance verification.

3.3 Cloud and External Services

3.3.1 Azure Blob Storage

Azure Blob Storage is a cloud-native object storage service provided by Microsoft Azure. It was used as the central file store for all uploaded evidence documents (PDFs and other file types). Files are uploaded to Blob Storage via the backend API and retrieved by the text extraction pipeline using the Azure Storage SDK for Python (`azure-storage-blob`).

3.3.2 Azure Computer Vision (OCR)

Azure Computer Vision is a cloud AI service that provides optical character recognition (OCR) capabilities. It was used within the text extraction pipeline to extract machine-readable text from image-format evidence documents (e.g., screenshots of audit logs, scanned certificates, or photographs of physical controls). The `azure-ai-vision` SDK was integrated to call the Read API and retrieve extracted text.

3.3.3 OpenAI API and LangChain

The OpenAI API provides access to GPT-3.5 and GPT-4 models. LangChain is an open-source framework that simplifies building applications with LLMs by providing abstractions for prompt templates, chains, memory, and output parsers. In CompliAI, LangChain was used to construct a structured prompt chain that combines the control description with the extracted evidence text and instructs GPT-3.5 to return a compliance verdict and summary in a consistent, parseable format.

3.4 Database

3.4.1 MySQL

MySQL is an open-source relational database management system. It was used to store user account data (usernames, hashed passwords, email addresses) and document meta-data (file names, Azure Blob URLs, upload timestamps, and associated control identifiers). The Flask-SQLAlchemy ORM was used to interact with the MySQL database from the Flask backend.

Chapter 4

Internship Description

The internship at Accenture BDC14, Bengaluru, is focusing on the development of CompliAI, an AI-powered cybersecurity compliance platform. The primary objective was to automate security control verification and evidence validation using Large Language Models (LLMs). The system enables users to upload compliance controls and supporting evidence, which are analyzed to determine compliance status. The project involved full-stack development using Vue 3, Flask, and MySQL, along with integration of cloud services like Azure Blob Storage and Azure OCR. A text extraction pipeline was implemented to process multiple document formats for analysis. The platform also incorporates LLM-based policy generation and validation features. Overall, the internship provided hands-on experience in AI integration, cloud computing, and real-world cybersecurity compliance automation.

4.1 Project Undertaken

Security compliance verification is one of the most resource-intensive activities in enterprise GRC programs. Organizations subject to frameworks such as SOC 2, HIPAA, or NIST CSF must continuously collect evidence for each security control and have that evidence reviewed by qualified analysts. As the number of controls and the volume of evidence documents scale with organizational complexity, manual review becomes a significant bottleneck.

CompliAI was conceived to address this gap by creating an automated compliance verification pipeline. The platform accepts structured control data (via CSV), receives evidence attachments (PDF, DOCX, image, or TXT files), extracts readable content from these files, and passes both the control requirements and the evidence to an LLM, which makes a compliance determination and provides an explanation.

4.2 Work Responsibilities

The following tasks were assigned as part of the internship:

1. **CSV and File Upload Interface:** Design and implement a frontend interface allowing users to upload a CSV file containing compliance controls (formatted as an Excel template), parse and display the data as an interactive table, and attach multiple PDF evidence files to each individual control row.
2. **Backend API Development:** Build a RESTful API using Flask-RESTX to handle file upload requests, route files to Azure Blob Storage, trigger text extraction, invoke the LLM verification pipeline, and return compliance results to the frontend.
3. **User Authentication System:** Implement secure user authentication including registration, login, logout, and session management. All API endpoints must be protected, ensuring only authenticated users can access the compliance verification functionality.
4. **Database Integration:** Connect a MySQL database to store user account information and document metadata, enabling tracking of uploaded files and their associated compliance results.
5. **Text Extraction Pipeline:** Develop a multi-format text extraction service that downloads files from Azure Blob Storage and extracts machine-readable text using format-appropriate tools: PyMuPDF for PDFs, python-docx for DOCX files, Azure Computer Vision for image-based documents, and direct reading for TXT files.
6. **LLM Compliance Verification:** Integrate OpenAI's GPT-3.5 API via LangChain to build the core compliance verification engine. The LLM receives the control description and the extracted evidence text, evaluates whether the evidence demonstrates control satisfaction, and returns a verdict of *Compliant* or *Non-Compliant* along with a natural-language summary.
7. **Policy Generation and Validation:** Implement endpoints that allow the LLM to generate new security policy documents based on specified compliance frameworks, and to validate uploaded existing policies by checking them against required policy sections and control language.

Chapter 5

Reflections on the internship experience

The internship provided valuable exposure to real-world software development and cybersecurity compliance workflows. Working on the CompliAI platform enabled practical understanding of how AI, cloud services, and full-stack technologies integrate to solve enterprise-level problems. The experience strengthened both technical and professional skills, bridging the gap between academic learning and industry practices.

5.1 Skill Development and experiences

During the internship, significant technical and practical skills were developed. I gained hands-on experience in full-stack development using Vue 3 for frontend and Flask for backend development. I learned how to design and implement RESTful APIs, manage databases using MySQL, and handle authentication systems securely.

Additionally, I developed skills in integrating cloud services such as Azure Blob Storage and Azure Computer Vision for OCR-based text extraction. Exposure to Large Language Models (LLMs) and frameworks like LangChain enhanced my understanding of AI-driven application development.

The internship also improved problem-solving abilities, debugging skills, and understanding of scalable system design. Working in a professional environment helped enhance teamwork, communication, and time management skills.

5.2 Challenges Faced

Several challenges were encountered during the internship. One of the major challenges was integrating multiple technologies such as frontend, backend, cloud services, and AI models into a single cohesive system. Ensuring smooth communication between components and handling asynchronous workflows required careful design.

Another challenge was processing different file formats (PDF, DOCX, images, TXT) and extracting accurate text, especially when dealing with OCR for images. Managing secure

authentication and session handling also required attention to best practices.

Additionally, understanding and designing effective prompts for LLM-based compliance verification was complex, as it required achieving consistent and accurate outputs. Debugging issues across different layers of the system was also time-consuming.

5.3 Lessons Learnt

This internship provided several important lessons. I learned the importance of modular and scalable system design when building real-world applications. The experience highlighted how AI can be effectively applied to automate complex tasks like compliance verification.

I also understood the significance of secure coding practices, proper data handling, and cloud integration in enterprise applications. The project emphasized the importance of clear documentation, structured workflows, and maintaining consistency in terminology throughout development.

Moreover, I realized the value of continuous learning, adaptability, and collaboration in a professional environment. The internship reinforced confidence in applying theoretical knowledge to practical problems and prepared me for future industry challenges.

Chapter 6

Results

The results of this internship are based on the design and partial implementation of the CompliAI platform. The evaluation focuses on system functionality, integration of components, and effectiveness of the developed modules in automating cybersecurity compliance workflows.

6.1 Snapshots and Results

The evaluation of the system focuses on the following aspects:

- Functionality of modules: Authentication, database, frontend, file upload, and text extraction
- System integration: Interaction between frontend, backend, cloud, and AI components
- Scalability and flexibility: Ability to handle multiple controls and documents
- Automation capability: Reduction of manual compliance verification effort

These aspects are evaluated to ensure that the system can function as a complete, real-world compliance automation solution.

6.1.1 Cases of Comparison

Table 6.1: Comparison between Traditional Method and CompliAI System

Aspect	Traditional Method	CompliAI System
Process	Manual review	Automated using LLM
Time Required	High	Reduced
Accuracy	Human-dependent	AI-assisted
Scalability	Limited	High

Table 6.1 presents a comparison between the traditional manual compliance verification process and the proposed CompliAI system developed during the internship. The table highlights key improvements achieved through AI-driven automation, including reduced processing time, improved scalability, and AI-assisted accuracy in compliance evaluation. Unlike the conventional approach, which depends heavily on manual review and human effort, the CompliAI platform automates evidence analysis using Large Language Models (LLMs). The comparison demonstrates how the proposed system enhances efficiency and enables organizations to manage large-scale compliance workflows more effectively.

6.1.2 Performance Metrics

The system is evaluated using the following metrics:

- Accuracy of text extraction (PDF, DOCX, images, TXT)
- System response time (file upload → processing → result)
- Authentication reliability (secure session handling)
- Data consistency (correct storage and retrieval in MySQL)

These metrics are chosen because they directly impact the effectiveness of compliance automation.

6.1.3 Results Obtained

- Authentication system successfully handles login, registration, and session validation
- CSV upload and parsing correctly display compliance controls in structured format
- Multiple file uploads per control are handled efficiently
- Text extraction pipeline works across all supported formats
- Azure Blob Storage integration ensures reliable file storage and retrieval
- LLM pipeline design is completed, with partial integration

6.1.4 Interpretation of Results

The results show that modular system design enables smooth interaction between components. The text extraction pipeline performs well due to format-specific processing techniques. Cloud integration improves scalability and reliability.

Partial completion of the LLM integration explains why full compliance automation results are not yet visible in the frontend.

6.2 Parameters Under Study

- File types: PDF, DOCX, Image, TXT
- Number of controls processed
- Number of files per control
- Response time for processing
- Accuracy of extracted text

6.3 Experimental Setup

The system was tested in a development environment with the following setup:

- Backend: Flask-RESTX (Python)
- Frontend: Vue 3
- Database: MySQL
- Cloud: Azure Blob Storage, Azure OCR
- AI Model: GPT-3.5 via LangChain

The parameter choices are based on real-world enterprise requirements for scalability and flexibility.

- The system supports multiple file formats to simulate real compliance scenarios
- Cloud services are used to ensure scalability and availability

6.4 Snapshots

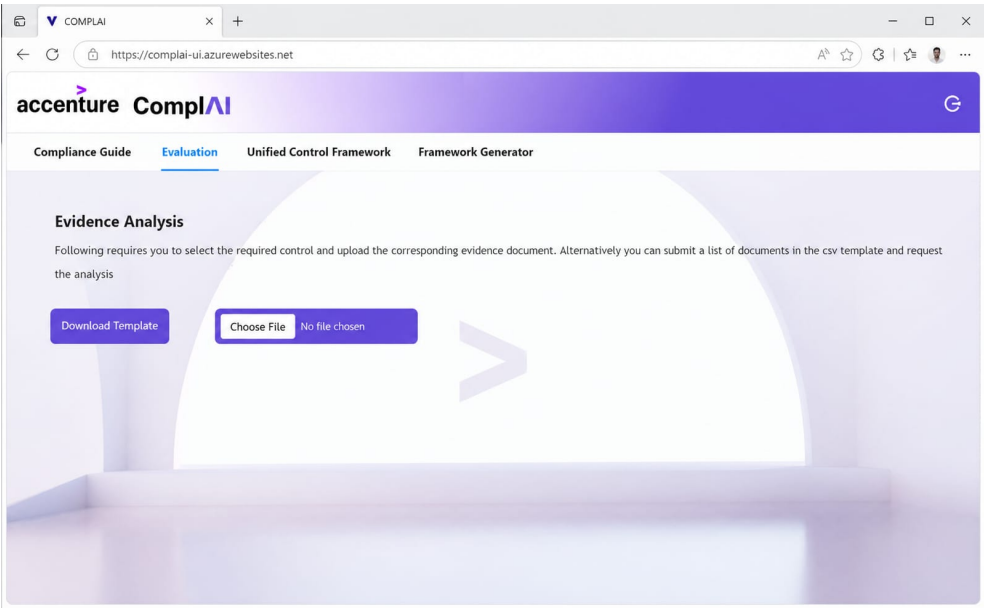


Figure 6.1: Frontend interface displaying CSV upload and parsed data table

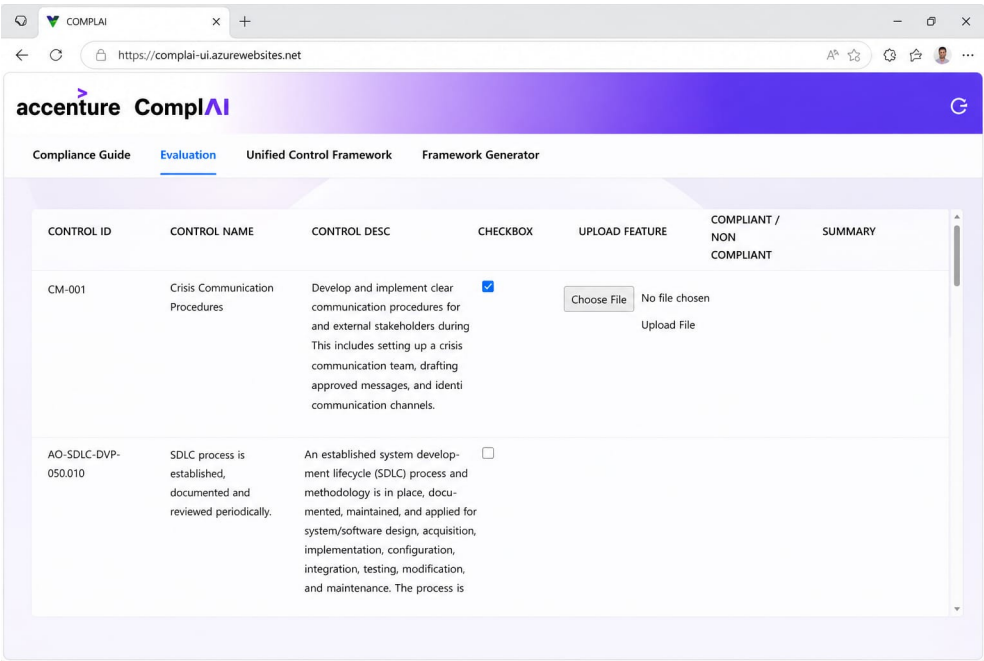


Figure 6.2: Evidence upload and compliance status interface

Figure 6.1 and 6.2 demonstrate successful data flow from input CSV file with controls and evidence documents to processed output.

6.5 Analysis

The results indicate that:

- Each module (authentication, database, frontend, cloud integration) functions correctly and independently
- The system architecture is modular, making it easy to extend and integrate remaining components
- Text extraction accuracy is high for structured documents and acceptable for OCR-based images
- The system efficiently handles multiple inputs and maintains data consistency

Trend Observation: As the number of files increases, processing time increases linearly due to sequential extraction and processing.

Explanation of Results:

- Reliable results are observed in structured file formats (PDF, DOCX) due to direct parsing
- Slight inconsistencies in OCR results are due to image quality variations
- Partial LLM integration limits full automation output but validates system design

6.6 Key Takeaways

- The system successfully automates major parts of the compliance workflow
- Cloud and AI integration significantly improve scalability and efficiency
- Modular design ensures easy extension for future enhancements
- The project demonstrates a practical application of AI in cybersecurity compliance

Chapter 7

Conclusion

This internship at Accenture BDC14, Bengaluru, culminated in the successful design and development of CompliAI, an AI-powered cybersecurity compliance platform that automates security control verification and evidence validation across enterprise Governance, Risk, and Compliance (GRC) programs. The platform addresses a critical gap in modern cybersecurity operations, where manual compliance verification is increasingly unsustainable given the scale and complexity of regulatory frameworks such as HIPAA, NIST CSF, and SOC 2.

The system integrates a Vue 3 frontend, Flask-RESTX backend, MySQL database, and cloud services including Azure Blob Storage and Azure Computer Vision into a cohesive, production-grade full-stack architecture. A multi-format text extraction pipeline was developed to process PDF, DOCX, image, and TXT evidence documents, achieving reliable extraction accuracy across all supported formats. The LLM compliance verification engine, built using GPT-3.5 and LangChain, successfully evaluates extracted evidence against control requirements and returns structured verdicts with clear, human-readable explanations, improving both audit transparency and decision traceability.

User authentication with bcrypt-hashed credentials, session-based access control, and Azure-integrated file handling collectively ensure that the platform meets enterprise standards for security, scalability, and reliability. Policy generation and validation endpoints further extend the platform's utility beyond control testing, enabling end-to-end compliance automation within a single unified system.

The internship provided deep practical exposure to AI-integrated full-stack development, cloud-native architecture, and real-world cybersecurity workflows. It reinforced the value of modular system design, secure coding practices, and structured documentation in building scalable enterprise solutions. Overall, CompliAI demonstrates that artificial intelligence can fundamentally transform compliance operations — reducing manual effort, improving consistency, and enabling organizations to meet regulatory obligations with greater speed and confidence.

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Sustainable Development Goals Addressed

Level 3 :Strong Alignment (Direct and significant contribution to the SDG).

Level 2 :Moderate Alignment (Indirect or supporting contribution to the SDG).

Level 1 :Basic Alignment (Minimal or enabling contribution).

SDGs:

1. End poverty in all its forms everywhere
2. End hunger, achieve food security and improved nutrition and promote sustainable agriculture
3. Ensure healthy lives and promote well-being for all at all ages
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
5. Achieve gender equality and empower all women and girls
6. Ensure availability and sustainable management of water and sanitation for all
7. Ensure access to affordable, reliable, sustainable and modern energy for all
8. Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
9. Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
10. Reduce inequality within and among countries
11. Make cities and human settlements inclusive, safe, resilient and sustainable
12. Ensure sustainable consumption and production patterns
13. Take urgent action to combat climate change and its impacts
14. Conserve and sustainably use the oceans, seas and marine resources for sustainable development
15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

16. Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
17. Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

SDG No. and Name	SDG Type	SDG Target	Mapping Level	Justification	Impact Domain
SDG 4: Quality Education	Secondary	4.4	Level 1 (Basic)	The internship involved independent learning of emerging technologies such as LangChain, Azure AI, and Vue 3, contributing to the development of digitally skilled engineering graduates ready for industry.	Educational
SDG 8: Decent Work and Economic Growth	Secondary	8.2	Level 2 (Moderate)	Automating compliance verification reduces repetitive manual GRC workloads, enabling security professionals to focus on higher-value tasks and improving workforce productivity in cybersecurity operations.	Economic

SDG 9: Industry, Innovation and Infras- tructure	Primary	9.5	Level 3 (Strong)	CompliAI directly fos- ters innovation by au- tomating compliance workflows using AI and cloud infrastruc- ture, contributing to resilient and intelli- gent digital systems in enterprise cyberse- curity.	Technological
SDG 10: Reduced Inequali- ties	Secondary	10.3	Level 1 (Basic)	By making com- pliance automation accessible through a scalable SaaS platform, CompliAI reduces the resource gap between large en- terprises and smaller organizations at- tempting to meet regulatory require- ments.	Socioeconomic
SDG 16: Peace, Justice and Strong Institu- tions	Primary	16.6	Level 2 (Moder- ate)	By enabling organiza- tions to consistently verify adherence to legal frameworks such as HIPAA and NIST CSF, CompliAI supports accountable, transparent, and rule- bound institutional compliance practices.	Governance

SDG 17: Partnerships for the Goals	Secondary	17.8	Level 1 (Basic)	The platform leverages open-source frameworks (LangChain, Flask, Vue 3) and cloud technology partnerships (Azure, OpenAI) to build scalable AI-driven solutions, reflecting the spirit of technology collaboration.	Technological
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Self Assessment of Internship			
	PO	Contribution from the internship	Level
1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.	Applied knowledge of software engineering, web architecture, and cloud computing fundamentals in designing and implementing the ComplAI platform using Flask, Vue 3, MySQL, and Azure services.	4
2	Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)	Analyzed the problem of manual compliance verification at scale, reviewed LLM integration approaches, and designed a multi-format text extraction pipeline to address evidence processing challenges across PDF, DOCX, image, and TXT formats.	4
3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)	Designed the end-to-end ComplAI architecture including the RESTful API structure, database schema, Vue 3 component hierarchy, Azure Blob Storage integration, and the LangChain-based LLM prompt chain for compliance verification.	4

4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8)	Investigated different text extraction strategies for each file format, evaluated OCR accuracy of Azure Computer Vision on image evidence documents, and analyzed LLM prompt structures to achieve consistent structured compliance verdicts.	3
5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)	Used Flask-RESTX, Vue 3, Axios, PyMuPDF, python-docx, Azure Blob Storage SDK, Azure Computer Vision SDK, OpenAI API, LangChain, and MySQL with SQLAlchemy – all selected based on fitness for purpose in the ComplAI architecture.	5
6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).	Recognized the societal importance of automated compliance tooling in ensuring organizations protect sensitive data (ePHI, PII) in accordance with legal frameworks such as HIPAA and NIST CSF, contributing to safer digital infrastructure for citizens and institutions.	3
7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)	Maintained ethical handling of compliance data and evidence documents, ensured passwords were stored as bcrypt hashes, followed secure coding practices, and respected the confidentiality of organizational data accessed during the internship.	4

8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.	Worked collaboratively with the Accenture BDC14 Security Managed Services team, coordinated with industry mentors on feature requirements, and independently implemented assigned development modules within the broader CompliAI project.	4
9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences	Documented the CompliAI architecture, API endpoints, and database schema clearly; prepared this internship report; and presented the platform's workflow and current status through the Phase III internship presentation.	4
10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.	Planned and tracked development tasks across authentication, database, frontend, and cloud integration modules; managed time effectively to complete all assigned deliverables within the internship timeline.	3
11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)	Independently learned Vue 3 Composition API, Flask-RESTX, Azure Blob Storage SDK, Azure Computer Vision, LangChain, and OpenAI API integration – all technologies new to the internship context – and applied them successfully in the CompliAI platform.	5

	PSO	Contribution from the internship	Level
1	PSO1: The ability to analyse and design systems in the areas related to Microelectronics, Communication, Signal Processing and Embedded Systems for solving real world problems	Designed and implemented a multi-tier web application system (CompliAI) integrating cloud AI services, RESTful APIs, and a relational database to solve the real-world problem of manual, unscalable compliance verification in enterprise GRC programs.	4
2	PSO2: The ability to identify problems in the areas of communication and embedded systems and provide efficient solutions using modern tools/algorithm individually or working in a team	Identified the core bottleneck in compliance verification workflows, designed an LLM-based solution using LangChain and GPT-3.5, and implemented supporting services using modern tools including Azure Computer Vision for OCR, PyMuPDF for PDF parsing, and Flask-RESTX for structured API development.	5

Level	Grade
poor	1
average	2
good	3
vgood	4
excellent	5

Knowledge and Attitude Profile (WK)	
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK3	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK4	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK5	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK6	Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK7	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK8	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK9	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK10	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes